THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 2^7	Add the exponents: $3 + 4 = 7$
2	**(d)** 2^5	Subtract the exponents: $9 - 4 = 5$
3	**(d)** 2^6	Multiply the exponents
4	**(b)** $\frac{1}{16}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{16}$
5	**(d)** 3.2×10^3	One non-zero digit before the decimal point
6	**(b)** 3400	Move the decimal point right for a positive power and left for a negative power
7	**(a)** 3^7	Add the exponents: $2 + 5 = 7$
8	**(c)** 3^5	Subtract the exponents: $8 - 3 = 5$
9	**(b)** 3^8	Multiply the exponents
10	**(c)** $\frac{1}{27}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{27}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Add exponents: 2^5 (1); $= 32$ (1).	1+1
12	$2^{-6} = \frac{1}{2^6}$ (1) $= \frac{1}{64}$ (1).	1+1
13	2.8×10^4 (2).	2
14	Add exponents: 3^4 (1); $= 81$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2^5 \times 2^4 = 2^9$ (1); $2^9 \div 2^6 = 2^3$ (1); value 8 (1).	1+1+1
16	$4.2 \times 10^5 = 420000$ (1); $3.5 \times 10^{-3} = 0.0035$ (1); move the decimal point right for a positive power and left for a negative power (1).	1+1+1
17	$73000000 = 7.3 \times 10^7$ (1); $0.0000064 = 6.4 \times 10^{-6}$ (1); one non-zero digit before the decimal point (1).	1+1+1
	OR: $3^4 \times 3^3 = 3^7$ (1); $3^7 \div 3^5 = 3^2$ (1); value 9 (1).	1+1+1
18	$7.1 \times 10^4 = 71000$ (1); $6.2 \times 10^{-2} = 0.062$ (1); move the decimal point right for a positive power and left for a negative power (1).	1+1+1
	OR: $4200000 = 4.2 \times 10^6$ (1); $0.0081 = 8.1 \times 10^{-3}$ (1); one non-zero digit before the decimal point (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 32 (1); (ii) $\frac{1}{9}$ (1); (iii) 1 (1); (iv) sum $= 32 + \frac{1}{9} + 1 = \frac{298}{9}$ (2).	1+1+1+2
	OR: $(2^2)^3 = 2^6$ (1); divide: 2^2 (1); multiply: 2^3 (2); value 8 (1).	1+1+2+1
20	(i) 27 (1); (ii) $\frac{1}{16}$ (1); (iii) 1 (1); (iv) sum $= 27 + \frac{1}{16} + 1 = \frac{449}{16}$ (2).	1+1+1+2
	OR: $(3^3)^2 = 3^6$ (1); divide: 3^2 (1); multiply: 3^4 (2); value 81 (1).	1+1+2+1

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5.8×10^4	One non-zero digit before the decimal point
2	**(d)** 650000	Move the decimal point right for a positive power and left for a negative power
3	**(b)** 5^5	Add the exponents: $3 + 2 = 5$
4	**(d)** 5^5	Subtract the exponents: $7 - 2 = 5$
5	**(a)** 5^6	Multiply the exponents
6	**(d)** $\frac{1}{125}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{125}$
7	**(a)** 4.7×10^5	One non-zero digit before the decimal point
8	**(d)** 0.027	Move the decimal point right for a positive power and left for a negative power
9	**(a)** 7^6	Add the exponents: $2 + 4 = 6$
10	**(d)** 7^4	Subtract the exponents: $10 - 6 = 4$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3^{-4} = \frac{1}{3^4}$ (1) $= \frac{1}{81}$ (1).	1+1
12	5.3×10^{-3} (2).	2
13	Add exponents: 5^3 (1); $= 125$ (1).	1+1
14	$7^{-2} = \frac{1}{7^2}$ (1) $= \frac{1}{49}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$5^3 \times 5^2 = 5^5$ (1); $5^5 \div 5^4 = 5^1$ (1); value 5 (1).	1+1+1
16	$2.9 \times 10^6 = 2900000$ (1); $8.4 \times 10^{-4} = 0.00084$ (1); move the decimal point right for a positive power and left for a negative power (1).	1+1+1
17	$560000 = 5.6 \times 10^5$ (1); $0.000092 = 9.2 \times 10^{-5}$ (1); one non-zero digit before the decimal point (1).	1+1+1
	OR: $7^2 \times 7^3 = 7^5$ (1); $7^5 \div 7^3 = 7^2$ (1); value 49 (1).	1+1+1
18	$5.3 \times 10^3 = 5300$ (1); $9.7 \times 10^{-5} = 0.000097$ (1); move the decimal point right for a positive power and left for a negative power (1).	1+1+1
	OR: $9100000000 = 9.1 \times 10^9$ (1); $0.00037 = 3.7 \times 10^{-4}$ (1); one non-zero digit before the decimal point (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 25 (1); (ii) $\frac{1}{16}$ (1); (iii) 1 (1); (iv) sum $= 25 + \frac{1}{16} + 1 = \frac{417}{16}$ (2).	1+1+1+2
	OR: $(5^2)^2 = 5^4$ (1); divide: 5^1 (1); multiply: 5^2 (2); value 25 (1).	1+1+2+1
20	(i) 64 (1); (ii) $\frac{1}{25}$ (1); (iii) 1 (1); (iv) sum $= 64 + \frac{1}{25} + 1 = \frac{1626}{25}$ (2).	1+1+1+2
	OR: $(2^3)^3 = 2^9$ (1); divide: 2^4 (1); multiply: 2^6 (2); value 64 (1).	1+1+2+1

Mistakes to avoid:

- Writing $2^{-3} = -8$ or $-\frac{1}{8}$; a negative exponent means the **reciprocal**, so $2^{-3} = \frac{1}{8}$.
- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
- Saying $a^0 = 0$; for $a \neq 0$ it is 1.
- Multiplying exponents when multiplying powers of the same base ($a^m \times a^n = a^{mn}$) or adding them in $(a^m)^n$.
- Leaving the answer as 25×10^4 or 0.25×10^6; in standard form the number part must satisfy $1 \leq m < 10$.
- Getting the sign of the exponent wrong for small numbers: $0.0003 = 3 \times 10^{-4}$, not 3×10^4.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 10^8	Multiply the exponents
2	**(b)** $\frac{1}{100}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{100}$
3	**(c)** 7.2×10^{-3}	One non-zero digit before the decimal point
4	**(c)** 0.00081	Move the decimal point right for a positive power and left for a negative power
5	**(c)** 4^8	Add the exponents: $3 + 5 = 8$
6	**(b)** 4^4	Subtract the exponents: $9 - 5 = 4$
7	**(b)** 4^9	Multiply the exponents
8	**(c)** $\frac{1}{64}$	$a^{-n} = \frac{1}{a^n} = \frac{1}{64}$
9	**(a)** 9.1×10^{-4}	One non-zero digit before the decimal point
10	**(a)** 6^5	Add the exponents: $2 + 3 = 5$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1.45×10^6 (2).	2
12	Add exponents: 10^5 (1); $= 100000$ (1).	1+1
13	$10^{-4} = \frac{1}{10^4}$ (1) $= \frac{1}{10000}$ (1).	1+1
14	2.1×10^{-4} (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$10^4 \times 10^5 = 10^9$ (1); $10^9 \div 10^6 = 10^3$ (1); value 1000 (1).	1+1+1
16	$1.6 \times 10^7 = 16000000$ (1); $4.8 \times 10^{-1} = 0.48$ (1); move the decimal point right for a positive power and left for a negative power (1).	1+1+1
17	$38000 = 3.8 \times 10^4$ (1); $0.00000025 = 2.5 \times 10^{-7}$ (1); one non-zero digit before the decimal point (1).	1+1+1
	OR: $4^2 \times 4^3 = 4^5$ (1); $4^5 \div 4^2 = 4^3$ (1); value 64 (1).	1+1+1
18	$3.8 \times 10^2 = 380$ (1); $2.5 \times 10^{-6} = 0.0000025$ (1); move the decimal point right for a positive power and left for a negative power (1).	1+1+1
	OR: $620000000 = 6.2 \times 10^8$ (1); $0.0043 = 4.3 \times 10^{-3}$ (1); one non-zero digit before the decimal point (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 36 (1); (ii) $\frac{1}{1000}$ (1); (iii) 1 (1); (iv) sum $= 36 + \frac{1}{1000} + 1 = \frac{37001}{1000}$ (2).	1+1+1+2
	OR: $(4^2)^3 = 4^6$ (1); divide: 4^2 (1); multiply: 4^3 (2); value 64 (1).	1+1+2+1
20	(i) 49 (1); (ii) $\frac{1}{32}$ (1); (iii) 1 (1); (iv) sum $= 49 + \frac{1}{32} + 1 = \frac{1601}{32}$ (2).	1+1+1+2
	OR: $(10^2)^3 = 10^6$ (1); divide: 10^2 (1); multiply: 10^4 (2); value 10000 (1).	1+1+2+1

Mistakes to avoid:

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- Adding the bases: $2^3 \times 2^4 = 4^7$ instead of $2^{3+4} = 2^7$.
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